INCH-POUND

The documentation and process conversion measures necessary to comply with this revision shall be completed by 7 June 2024.

MIL-PRF-19500/452K
w/AMENDMENT 1
7 February 2024
SUPERSEDING
MIL-PRF-19500/452K
29 January 2019

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, DIODE, SILICON, VOLTAGE-REFERENCE, TEMPERATURE COMPENSATED, TYPES 1N4565A THROUGH 1N4584A, LEADED AND SURFACE MOUNT PACKAGES AND UNENCAPSULATED DIE, QUALITY LEVELS JAN, JANTX, JANTXV, JANS, JANHC, AND JANKC, RADIATION HARDENED (TOTAL DOSE ONLY)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for 6.4 volts ±5 percent, silicon, low bias current, temperature compensated, voltage-reference diodes. Four levels of product assurance (JAN, JANTX, JANTXV, and JANS) are provided for each encapsulated device type as specified in MIL-PRF-19500, and two levels of product assurance for each unencapsulated device type die. Seven levels of radiation hardened (total dose only) assurance are provided for quality levels JANTXV, JANS, JANHC and JANKC as specified in MIL-PRF-19500.

1.2 Physical dimensions.

1.2.1 Package outlines. The device package outlines are as follows: axial leaded packages DO-204AA or DO-204AH (similar to DO-7 and DO-35) in accordance with figure 1 and surface mount version metal electrode leadless face (MELF) (similar to DO-213AA) in accordance with figure 2.

1.2.2 Unencapsulated die topography The dimensions and topography for unencapsulated die are in accordance with figure 3 (JANHC and JANKC), figure 4 (JANHCB and JANKCB) figure 5 (JANHCC and JANKCC), and figure 6 (JANHCD and JANKCD).

1.3 Maximum ratings. Unless otherwise specified T_A = +25°C.

P_T	T_{STG} and T_J	I_{ZM} (1)	Power derating above T_A = +25°C
mW	°C	mA dc	mW/°C
500	-55 to +175	70	3.33

(1) To guarantee voltage temperature stability, it is necessary to maintain the proper I_Z as specified in 1.4 herein.

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1.4 Primary electrical characteristics. Unless otherwise specified, at T_A = +25°C.

Column 1	Column 2	Column 3	Column 4		Column 5
Type (1) I_Z = 0.5 mA dc	ΔV_Z	Z_Z (2)	V_Z (2)		I_R V_R = 3.0 V
			Min	Max	
	V dc	ohms	Volts	Volts	µA
1N4565A-1, 1N4565AUR-1	0.100	200	6.08	6.72	2.0
1N4566A-1, 1N4566AUR-1	0.050	200	6.08	6.72	2.0
1N4567A-1, 1N4567AUR-1	0.020	200	6.08	6.72	2.0
1N4568A-1, 1N4568AUR-1	0.010	200	6.08	6.72	2.0
1N4569A-1, 1N4569AUR-1	0.005	200	6.08	6.72	2.0
Type (1) I_Z = 1.0 mA dc	ΔV_Z	Z_Z (2)	V_Z (2)		I_R V_R = 3.0 V
			Min	Max	
	V dc	ohms	Volts	Volts	µA
1N4570A-1, 1N4570AUR-1	0.100	100	6.08	6.72	2.0
1N4571A-1, 1N4571AUR-1	0.050	100	6.08	6.72	2.0
1N4572A-1, 1N4572AUR-1	0.020	100	6.08	6.72	2.0
1N4573A-1, 1N4573AUR-1	0.010	100	6.08	6.72	2.0
1N4574A-1, 1N4574AUR-1	0.005	100	6.08	6.72	2.0
Type (1) I_Z = 2.0 mA dc	ΔV_Z	Z_Z (2)	V_Z (2)		I_R V_R = 3.0 V
			Min	Max	
	V dc	ohms	Volts	Volts	µA
1N4575A-1, 1N4575AUR-1	0.100	50	6.08	6.72	2.0
1N4576A-1, 1N4576AUR-1	0.050	50	6.08	6.72	2.0
1N4577A-1, 1N4577AUR-1	0.020	50	6.08	6.72	2.0
1N4578A-1, 1N4578AUR-1	0.010	50	6.08	6.72	2.0
1N4579A-1, 1N4579AUR-1	0.005	50	6.08	6.72	2.0
Type (1) I_Z = 4.0 mA dc	ΔV_Z	Z_Z (2)	V_Z (2)		I_R V_R = 3.0 V
			Min	Max	
	V dc	ohms	Volts	Volts	µA
1N4580A-1, 1N4580AUR-1	0.100	25	6.08	6.72	2.0
1N4581A-1, 1N4581AUR-1	0.050	25	6.08	6.72	2.0
1N4582A-1, 1N4582AUR-1	0.020	25	6.08	6.72	2.0
1N4583A-1, 1N4583AUR-1	0.010	25	6.08	6.72	2.0
1N4584A-1, 1N4584AUR-1	0.005	25	6.08	6.72	2.0

(1) Electrical characteristics and test conditions for "A-1" and "AUR-1" devices are identical.
(2) To guarantee ΔV_Z (voltage-temperature stability), it is necessary to maintain the proper I_Z as specified herein.

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1.5 Part or Identifying Number (PIN). The PIN is in accordance with MIL-PRF-19500, and as specified herein. See 6.4 for PIN construction example and 6.5 for a list of available PINs, and 6.6 for supersession information.

1.5.1 JAN certification mark and quality level.

1.5.1.1 Quality level designators for encapsulated devices. The quality level designators for encapsulated devices that are applicable for this specification sheet from the lowest to the highest level are as follows: "JAN", "JANTX", "JANTXV", and "JANS".

1.5.1.2 Quality level designators for unencapsulated devices (die). The quality level designators for unencapsulated devices that are applicable for this specification sheet are "JANHC" and "JANKC".

1.5.2 Device type. The designation system for the devices covered by this specification sheet is as follows.

1.5.2.1 First number and first letter symbols. The devices of this specification sheet use the first number and letter symbols "1N".

1.5.2.2 Second number symbols. The second number symbols for the devices covered by this specification sheet are as follows: "4565" through "4584".

1.5.3 Suffix symbols. The following suffix symbol(s) are incorporated into the PINs for this specification sheet.

1.5.3.1 Device modifier. The "A" suffix symbol indicates a modified version of the non-suffix device.

1.5.3.2 Package designators. The suffix symbols (or lack thereof) that designate the package outline for the devices covered by this specification sheet are as follows.

	A blank designator identifies that the package is a DO-204AA or DO-204AH (formerly DO-7or DO-35) in accordance with figure 1.
UR	A "UR" designator identifies that the package is a DO-213AA surface mount package in accordance with figure 2.

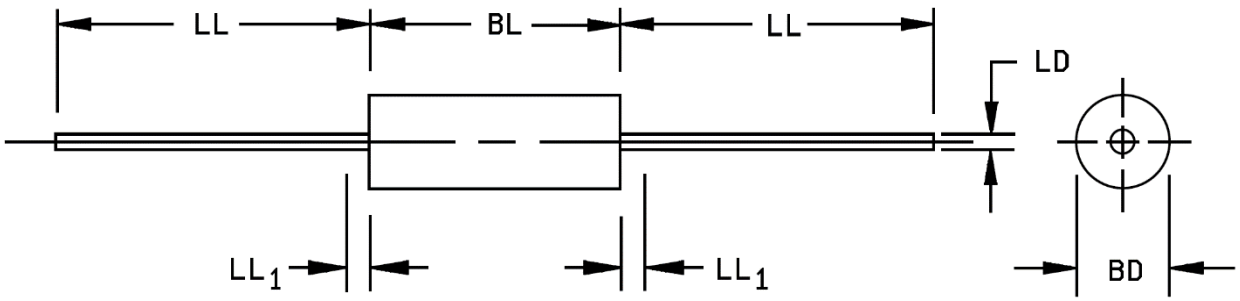
1.5.3.3 Metallurgical bond. The last suffix symbol "-1" designates the diode construction (see 3.4.2).

1.5.4 Radiation hardness assurance (RHA) designator. For quality levels JANTXV, JANS, JANHC and JANKC, the RHA levels that are applicable for this specification sheet from lowest to highest are as follows: "M", "D", "L", "R", "F", "G", and "H".

1.5.5 Lead finish. The lead finishes applicable to this specification sheet are listed on QPDSIS-19500.

1.5.6 Die identifiers for unencapsulated devices. The manufacturer die identifiers that are applicable for this specification sheet are "A", "B", "C" and "D" (see figures 3, 4, 5, 6 and 6.7).

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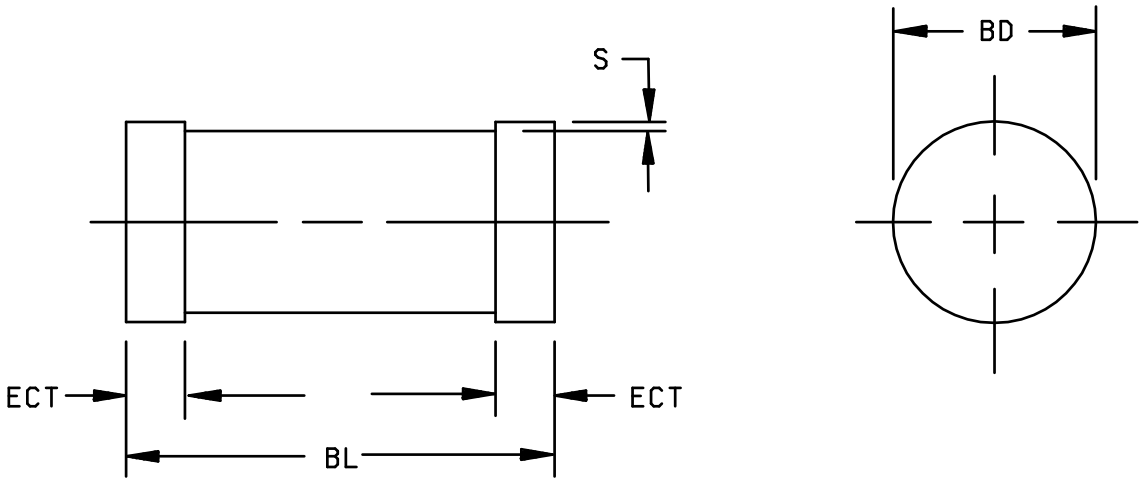


Symbol	Dimensions				Notes
	Inch	Inch	mm	mm	
	Min	Max	Min	Max	
BD	.060	.107	1.52	2.72	2
BL	.120	.300	3.05	7.62	2
LD	.018	.023	0.46	0.58	3
LL	1.000	1.500	25.40	38.10	
LL_1		0.050		1.27	4

NOTES:
1. Dimensions are in inches. Millimeters are given for general information only.
2. Package contour optional within dimension BD and length dimension BL. Heat slugs, if any, shall be included within this cylinder but shall not be subject to minimum limit of dimension BD.
3. Dimensions are pre-solder dip.
4. Within this zone, lead diameter may vary to allow for lead finishes and irregularities, other than heat slugs.

FIGURE 1. Physical dimensions of DO-204AA or DO-204AH (1N4565A-1 through 1N4584A-1).

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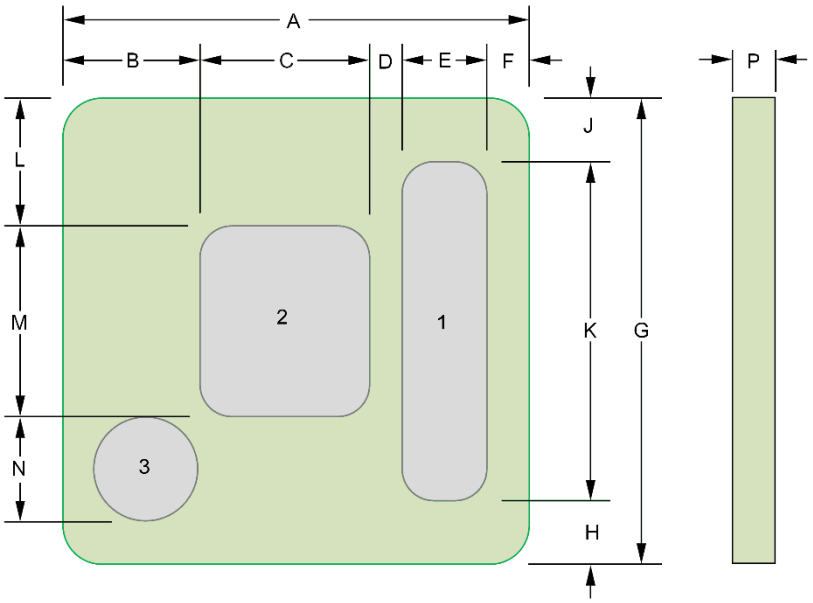
Symbol	Dimensions			
	Inch	Inch	mm	mm
	Min	Max	Min	Max
BD	.063	.067	1.60	1.70
ECT	.016	.022	0.41	0.56
BL	.130	.146	3.30	3.71
S	.001 Min		0.03 Min	

NOTES:
1. Dimensions are in inches. Millimeters are given for general information only.
2. Dimensions are pre-solder dip.
3. The S dimension is minimum clearance of glass body to mounting surface on all orientations.

FIGURE 2. Physical dimensions of DO-213AA (1N4565AUR-1 through 1N4584AUR-1).

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Symbol	Dimensions			
	inch	inch	mm	mm
	Min	Max	Min	Max
A	.0280	.0320	.711	.813
B	.0080	.0100	.203	.254
C	.0104	.0106	.264	.269
D	.0019	.0021	.048	.053
E	.0054	.0056	.137	.142
F	.0020	.0040	.050	.102
G	.0280	.0320	.711	.813
H	.0030	.0050	.076	.127
J	.0030	.0050	.076	.127
K	.0209	.0211	.531	.536
L	.0080	.0100	.203	.254
M	.0104	.0106	.264	.269
N	.0059	.0061	.150	.155
P	.0080	.0120	.203	.302



Design data

Circuit layout data:

Backside must be electrically isolated to ensure proper performance.

For Zener operation, cathode must be operated positive with respect to anode.

Metallization:

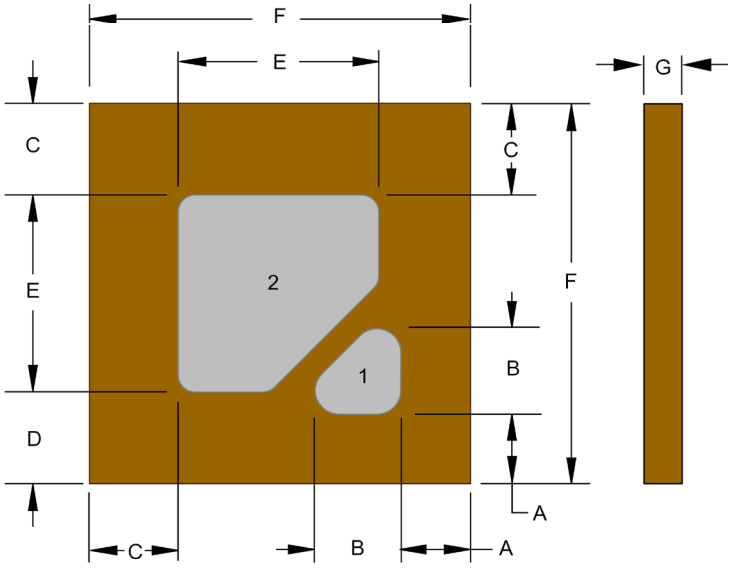
Top:	1 (Cathode)	Al	
	2 (Anode)	Al	
	3 (Test pad)	Al	See note 2.
Back:		Au	
Al thickness		25,000Å minimum.	
Au thickness		4,000Å minimum.	

NOTES:
1. Dimensions are in inches unless otherwise indicated. Millimeters are given for general information only.
2. Test pad is for wire bond evaluation only. No electrical contact is made with test pad.

FIGURE 3. Physical dimensions and characteristics for JANHC and JANKC (A-version) die.

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Symbol	Dimensions			
	inch	inch	mm	mm
	Min	Max	Min	Max
A	.0035	.0065	.088	.165
B	.0050	.0080	.127	.203
C	.0050	.0065	.127	.165
D	.0050	.0065	.127	.165
E	.0150	.0165	.381	.419
F	.0260	.0290	.660	.737
G	.0080	.0120	.203	.302



Design data

Circuit layout data:

Backside must be electrically isolated to ensure proper performance.

For zener operation, cathode must be operated positive with respect to anode.

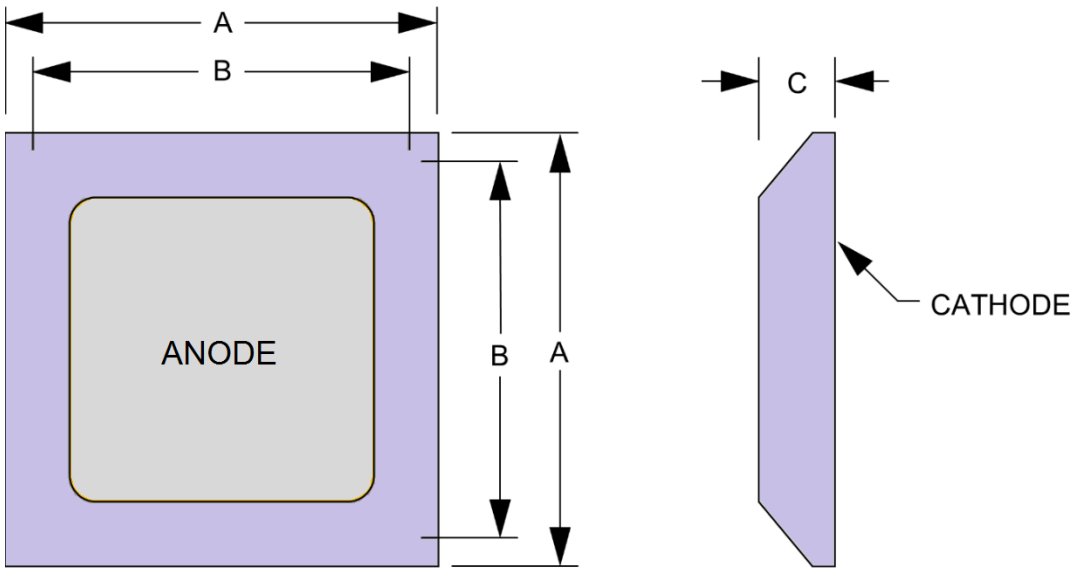
Metallization:

Top:	1 (Cathode)	Al
	2 (Anode)	Al
Back:		Au
Al thickness		40,000Å minimum.
Au thickness		5,000Å minimum.

NOTES:
1. Dimensions are in inches. Millimeters are given for general information.

FIGURE 4. Physical dimensions and characteristics for JANHC and JANKC (B-version) die.

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Symbol	Dimensions			
	Inch	Inch	mm	mm
	Min	Max	Min	Max
A	.024	.037	0.61	0.94
B	.014	.029	0.36	0.74
C	.008	.012	0.20	0.30

Design data

Circuit layout data:

For zener operation, cathode must be operated positive with respect to anode.

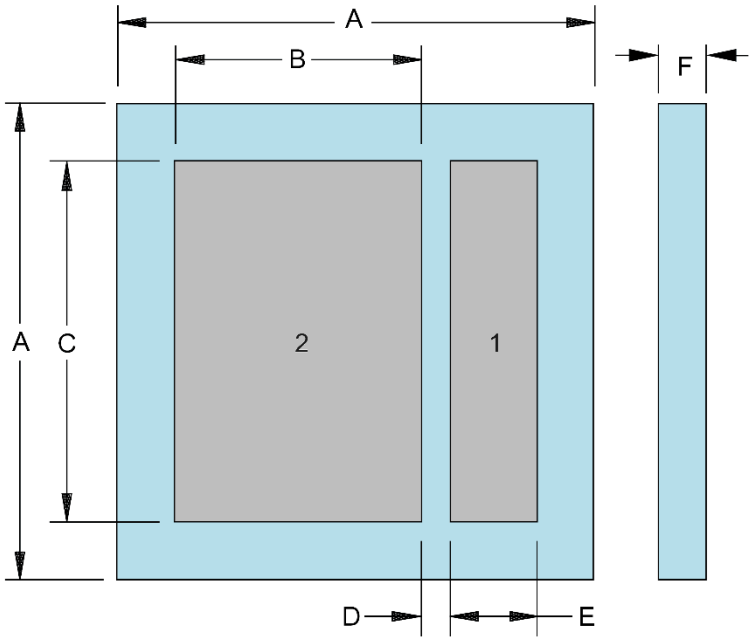
Metallization:

Top (Anode)	Al
Back (Cathode)	Au
Al thickness	40,000Å minimum.
Au thickness	5,000Å minimum.

NOTES:
1. Dimensions are in inches. Millimeters are given for general information.

FIGURE 5. Physical dimensions and characteristics for JANHC and JANKC (C-version) die.

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Symbol	Dimensions			
	Inch	Inch	mm	mm
	Min	Max	Min	Max
A	.031	.035	0.787	0.889
B	.016	.018	0.406	0.457
C	.024	.026	0.610	0.660
D	.001	.003	0.025	0.076
E	.005	.007	0.127	0.178
F	.008	.012	0.203	0.305

Design data

Circuit layout data: Not applicable.

Metallization:

Top	1 (Cathode)	Al
	2 (Anode)	Al
Back		Au
Al thickness		37,500Å minimum.
Au thickness		3,750Å minimum.

NOTES:
1. Dimensions are in inches. Millimeters are given for general information only.

FIGURE 6. Physical dimensions and characteristics for JANHC and JANKC (D-version) die.

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 – Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 – Test Methods For Semiconductor Devices.

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.4 Interface and physical dimensions. The interface and physical dimensions shall be as specified in MIL-PRF-19500 and figures 1, 2, 3, 4, 5, and 6 herein.

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4.2 Encapsulated diode construction. These devices shall be constructed to comply with the dash-one construction requirements of MIL-PRF-19500 utilizing either a category I or II metallurgical bond as defined in MIL-PRF-19500.

3.4.3 JANHC and JANKC construction. JANHC and JANKC construction may differ in die size and bonding pad layout provided the manufacturing technology is identical (example: Diffused junction, alloy junction).

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3.4.4 Axial lead package outline. This specification contains two standard axial lead packages; DO-204AA and DO-204AH (formerly DO-7 and DO-35). Any user of this specification that has a specific package outline requirement shall specify their preference in the document order. If package is not specified, the manufacturer may supply either package (see 6.2).

3.5 Marking. Devices shall be marked as specified in MIL-PRF-19500 and herein.

3.5.1 Axial lead devices. Manufacturer's identification and date code shall be marked on the devices The polarity shall be indicated with a contrasting color band to denote the cathode end. The prefixes JAN, JANTX, JANTXV, and JANS may be abbreviated as J, JX, JV, and JS respectively. The PIN may be reduced to J4565, JX4565, JV4565, or JS4565. Color coding shall not be permitted for part numbering.

3.5.2 Marking of "UR" version devices. For "UR" suffix devices only, all marking (except polarity) may be omitted from the body, but shall be retained on the initial container.

3.5.3 Polarity. The polarity shall be indicated with a contrasting color band to denote the cathode end. Alternately, for UR suffix devices, a minimum of three contrasting color dots spaced evenly around the periphery at the cathode end may be used.

3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I.

3.7 Electrical test requirements. The electrical test requirements shall be the subgroups specified in 4.4.2 and 4.4.3.

3.8 RHA requirements. The requirements and test levels for RHA shall be as defined in MIL-PRF-19500 and herein.

3.9 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4).

d. Element evaluation of un-encapsulated die (see 4.6).

4.1.1 Sampling inspection. Sampling inspection shall be in accordance with MIL-PRF-19500 and as specified herein, except that lot accumulation period shall be 3 months in lieu of 6 weeks.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.2.1 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not require the performance of table III tests, the tests specified in table III herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

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4.2.2 JANHC and JANKC devices. Qualification for JANHC and JANKC devices shall be in accordance with MIL-PRF-19500.

4.2.3 Radiation hardened devices. See MIL-PRF-19500 and 4.4.4 herein.

4.3 Screening of encapsulated devices (Quality levels JANS, JANTXV and JANTX only). Screening shall be in accordance with table E-IV of MIL-PRF-19500, and as specified herein. The following electrical measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen	Measurements for JANS level	Measurements for JANTXV and JANTX level
3b	Not applicable	Not applicable
3c	Not applicable	Not applicable
4	Not applicable	Not applicable
5	Not applicable	Not applicable
6	Not applicable	Not applicable
8	Required	Not required
9	Required	Not applicable
10	Not applicable	Not applicable
11	Required V_Z, Z_Z	Required V_Z, Z_Z
12	Required, see 4.3.1.1	Required, see 4.3.1.1
13	Required Subgroups 2 and 3 of table I herein; $\Delta Z_Z \leq \pm 15$ percent of initial reading $\Delta V_Z \leq \pm 0.005$ V dc from initial value at T_A = +25°C ±2°C	Required Subgroup 2 of table I herein; $\Delta Z_Z \leq \pm 15$ percent of initial reading. $\Delta V_Z \leq \pm 0.005$ V dc from initial value at T_A = +25°C ±2°C

4.3.1.1 Power burn-in conditions. Power burn-in conditions shall be as follows:

a. 1N4565 through 1N4569: I_Z = 0.5 mA dc ±0.05 mA dc, T_A = +150 +5°C, -0°C.

b. 1N4570 through 1N4574: I_Z = 1.0 mA dc ±0.1 mA dc, T_A = +150 +5°C, -0°C

c. 1N4575 through 1N4579: I_Z = 2.0 mA dc ±0.2 mA dc, T_A = +150 +5°C, -0°C.

d. 1N4580 through 1N4584: I_Z = 4.0 mA dc ±0.4 mA dc, T_A = +150 +5°C, -0°C.

To better utilize burn-in equipment, higher values of I_Z shall be permitted provided the junction temperature does not exceed +175°C.

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4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with table E-V of MIL-PRF-19500 and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with 4.4.2.1 for quality level JANS and 4.4.2.2 for quality levels JAN, JANTX, and JANTXV.

4.4.2.1 Quality level JANS. Group B inspection for quality level JANS shall be conducted in accordance with the conditions specified for subgroup testing in table E–VIA of MIL-PRF-19500 and herein. For purposes of JANS inspection, a single device type shall be defined as devices from a single wafer lot (for each die type used in the construction). The conformance inspection sample shall be selected from the part category with the lowest ΔV_Z rating in the inspection lot.

Subgroup	Method	Conditions
B3	1056	Test condition A, 25 cycles.
B3	4066	Not applicable.
B3	1071	Test condition E.
B4	1037	I_Z = 35 mA dc at T_A = room ambient; t_{on} = t_{off} = 30 seconds minimum for 4,000 cycles.
B5	1027	I_{ZM} = 70 mA dc for 96 hours. T_A = +75°C or adjusted as required, to give an average lot T_J = +200°C.
B6		Not applicable.

4.4.2.2 Quality levels JAN, JANTX and JANTXV. Group B inspection for quality levels JAN, JANTX, and JANTXV shall be conducted in accordance with the conditions specified in table E–VIB of MIL-PRF-19500 and herein.

Subgroup	Method	Conditions
B2	1056	Test condition A, 25 cycles.
B2	4066	Not applicable.
B2	1071	Test condition E.
B3	1027	Conditions for "A-1" and "AUR-1" versions (see 4.3.1.1).
B5		Not applicable.

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4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VII of MIL-PRF-19500.

Subgroup	Method	Conditions
C2	1056	Test condition A, 25 cycles.
C2	2036	Tension: Test condition A; 10 pounds weight (4.54 kg), t = 15 ±3 seconds; Lead fatigue: Test condition E Tension and lead fatigue test is not applicable to surface mount "UR" version devices.
C2	1071	Test condition E.
C3		Not applicable.
C5		Not applicable.
C6	1026	See 4.5.2. Conditions for "A-1" and "AUR-1" versions of: 1N4565 through 1N4569: I_Z = 0.5 mA dc, T_A = +100°C. 1N4570 through 1N4574: I_Z = 1.0 mA dc, T_A = +100°C. 1N4575 through 1N4579: I_Z = 2.0 mA dc, T_A = +100°C. 1N4580 through 1N4584: I_Z = 4.0 mA dc, T_A = +100°C.
C7		Not applicable.

4.4.4 Group D inspection. Group D inspection shall be conducted in accordance with table E-VIII of MIL-PRF-19500 and table II herein. Submitted lots for group D sample inspection must be constructed using one homogeneous wafer lot for the zener and one wafer lot for the compensating die (die), as also described in the Design and Construction form 36D (see table II herein) submitted to the qualifying activity.

4.4.5 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup tests in table E-IX of MIL-PRF-19500 and table III herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables as follows.

4.5.1 Reference voltage. The reference voltage test shall be performed in accordance with test method 4022 of MIL-STD-750. The test current shall be applied until thermal equilibrium is attained (15 s maximum) prior to reading the reference voltage. For this test, the diode shall be suspended by its leads with mounting clips whose inside edge is located at .375 inch (9.53 mm) from the body and the mounting clips shall be maintained at the specified temperature. This measurement may be performed after a shorter time following application of the test current than that which provide thermal equilibrium if correlation to stabilized readings can be established to the satisfaction of the Government.

4.5.2 Reference voltage time stability. The breakdown voltage shall be measured in accordance with 4.5.1 prior to life testing, at 340 hours, and at the conclusion of the life test. The 340-hour reading shall be compared with the 0-hour reading and the 1,000-hour reading compared with the 340-hour reading. The change in breakdown voltage shall not exceed the limits specified. The test temperature for breakdown voltage shall be the same as the specified ambient life test temperature (see table IV herein).

4.5.3 Voltage-temperature stability. The breakdown voltage of each diode type shall be measured in accordance with 4.5.1 and recorded at each of the specified temperatures. The lowest measured voltage shall be subtracted from the highest measured voltage for each diode. The difference value obtained shall not exceed the specified ΔV_Z for each diode type.

4.6 Element evaluation of unencapsulated die. The element evaluation of unencapsulated die shall be in accordance with appendix G of MIL-PRF-19500.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limit	Limit	Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2 2/						
Reference voltage	4022	See 4.5.1	V_Z	6.08	6.72	
1N4565 through 1N4569		I_Z = 0.5 ±0.01 mA dc				V dc
1N4570 through 1N4574		I_Z = 1.0 ±0.01 mA dc				V dc
1N4575 through 1N4579		I_Z = 2.0 ±0.01 mA dc				V dc
1N4580 through 1N4584		I_Z = 4.0 ±0.01 mA dc				V dc
Small signal breakdown impedance	4051		Z_Z			ohms
1N4565 through 1N4569		I_Z = 0.5 ±0.01 mA dc, I_{sig} = 0.05 mA rms			200	
1N4570 through 1N4574		I_Z = 1.0 ±0.01 mA dc, I_{sig} = 0.1 mA rms			100	
1N4575 through 1N4579		I_Z = 2.0 ±0.01 mA dc, I_{sig} = 0.2 mA rms			50	
1N4580 through 1N4584		I_Z = 4.0 ±0.01 mA dc, I_{sig} = 0.4 mA rms			25	
Subgroup 3						
Voltage-temperature stability (see 4.5.3)	4022	T_A = -55°C, 0°C, +25°C, +75°C, +100°C ±2°C, I_Z = column 4 of 1.4	ΔV_Z			
1N4565A-1, 1N4565AUR-1					100	mV dc
1N4570A-1, 1N4570AUR-1					100	mV dc
1N4575A-1, 1N4575AUR-1					100	mV dc
1N4580A-1, 1N4580AUR-1					100	mV dc
1N4566A-1, 1N4566AUR-1					50	mV dc
1N4571A-1, 1N4571AUR-1					50	mV dc
1N4576A-1, 1N4576AUR-1					50	mV dc
1N4581A-1, 1N4581AUR-1					50	mV dc

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit	Limit	Unit
	Method	Conditions		Min	Max	
Subgroup 3 - Continued		T_A = -55°C, 0°C, +25°C, +75°C, +100°C ±2°C, I_Z = column 4 of 1.4	ΔV_Z			
1N4567A-1, 1N4567AUR-1					20	mV dc
1N4572A-1, 1N4572AUR-1					20	mV dc
1N4577A-1, 1N4577AUR-1					20	mV dc
1N4582A-1, 1N4582AUR-1					20	mV dc
1N4568A-1, 1N4568AUR-1					10	mV dc
1N4573A-1, 1N4573AUR-1					10	mV dc
1N4578A-1, 1N4578AUR-1					10	mV dc
1N4583A-1, 1N4583AUR-1					10	mV dc
1N4569A-1, 1N4569AUR-1					5	mV dc
1N4574A-1, 1N4574AUR-1					5	mV dc
1N4579A-1, 1N4579AUR-1					5	mV dc
1N4584A-1, 1N4584AUR-1					5	mV dc
Subgroups 4, 5, and 6						
Not applicable						
Subgroup 7						
Reverse leakage current	4016	Condition A or B; V_R = 3.0 Vdc	I_R		2.0	µA

1/ For sampling plan, see MIL-PRF-19500.
2/ Electrical characteristics apply to all package styles.

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TABLE II. Group D inspection.

Inspection 1/ 2/	MIL-STD-750		Symbol	Pre-irradiation limits 3/		Post-irradiation limits 3/		ΔV_{Z1} 4/		Unit
	Method	Conditions		Min	Max	Min	Max	Min	Max	
Subgroup 1										
Not applicable										
Subgroup 2		T_C = +25°C								
Steady-state total dose irradiation	1019	I_Z = column 4 of 1.4, Condition A								
Reference voltage	4022	I_Z = column 4 of 1.4, (see 4.5.1)	V_Z	6.08	6.72	6.08	6.72			V dc
Small-signal breakdown impedance	4051	I_Z = column 4 of 1.4	Z_Z							ohms
1N4565					200		200			
1N4566		I_{sig} = 0.05 mA ac			200		200			
1N4567		I_{sig} = 0.05 mA ac			200		200			
1N4568		I_{sig} = 0.05 mA ac			200		200			
1N4569		I_{sig} = 0.05 mA ac			200		200			
1N4570		I_{sig} = 0.1 mA ac			100		100			
1N4571					100		100			
1N4572					100		100			
1N4573					100		100			
1N4574					100		100			
1N4575		I_{sig} = 0.2 mA ac			50		50			
1N4576					50		50			
1N4577					50		50			
1N4578					50		50			
1N4579					50		50			
1N4580		I_{sig} = 0.4 mA ac			25		25			
1N4581					25		25			
1N4582					25		25			
1N4583					25		25			
1N4584					25		25			

See footnotes at end of table.

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TABLE II. Group D inspection - Continued.

Inspection 1/ 2/	MIL-STD-750		Symbol	Pre-irradiation limits 3/		Post-irradiation limits 3/		ΔV_{Z1} 4/		Unit
	Method	Conditions		Min	Max	Min	Max	Min	Max	
Subgroup 2 - Continued.										
Reverse current leakage	4016	Condition A or B; V_R = 3.0 V dc	I_R		2.0		2.0			µA
Voltage stability	4022	I_Z = Column 4 of 1.4, T_A = +25°C ±2°C (see 4.5.3)	ΔV_Z							mV
1N4565									±3	
1N4566									±3	
1N4567									±2	
1N4568									±1.5	
1N4569									±1	
1N4570									±3	
1N4571									±3	
1N4572									±2	
1N4573									±1.5	
1N4574									±1	
1N4575									±3	
1N4576									±3	
1N4577									±2	
1N4578									±1.5	
1N4579									±1	
1N4580									±3	
1N4581									±3	
1N4582									±2	
1N4583									±1.5	
1N4584									±1	

1/ For sampling plan, see MIL-PRF-19500.
2/ Electrical characteristics and test conditions for "A-1" and "AUR-1" devices are identical.
3/ The limits listed are for quality levels JANS and JANTXV of RHA levels "M", "D", "L", "R", "F", "G", and "H".
4/ Pre-irradiation to post-irradiation change.

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TABLE III. Group E inspection qualification and requalification (all product assurance levels).

Inspection	MIL-STD-750		Sampling plan
	Method	Conditions	
Subgroup 1			45 devices, c = 0
Temperature cycling	1051	500 cycles.	
Electrical measurements		See table I, subgroup 2 herein.	
Subgroup 2			45 devices, c = 0
Steady-state operation life	1038	Condition B, 1,000 hours. (see 4.3.1).	
Electrical measurements		See table I, subgroup 2 herein.	
Subgroups 4, 5, 6, and 7			
Not applicable			
Subgroup 8			45 devices
Resistance to glass cracking	1057	Step stress to destruction by increasing cycles or up to a maximum of 25 cycles.	

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TABLE IV. Reference voltage time stability.

Step	Inspection	MIL-STD-750		Symbol	Limit	Limit	Unit
		Method	Condition		Min	Max	
1	Reference-voltage time stability	4022	T_A = +100°C ±2°C, 0 to 340 hours (see 4.5.2)	ΔV_Z			mV dc
	1N4565, 1N4570					10	
	1N4575, 1N4580					10	
	1N4566, 1N4571					8	
	1N4576, 1N4581					8	
	1N4567, 1N4572					7	
	1N4577, 1N4582					7	
	1N4568, 1N4573					6	
	1N4578, 1N4583					6	
	1N4569, 1N4574					5	
	1N4579, 1N4584					5	
2	Reference-voltage time stability	4022	T_A = +100°C ±2°C, 340 to 1,000 hours (see 4.5.2)	ΔV_Z			mV dc
	1N4565, 1N4570					5	
	1N4575, 1N4580					5	
	1N4566, 1N4571					4	
	1N4576, 1N4581					4	
	1N4567, 1N4572					4	
	1N4577, 1N4582					4	
	1N4568, 1N4573					3	
	1N4578, 1N4583					3	
	1N4569, 1N4574					3	
	1N4579, 1N4584					3	

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

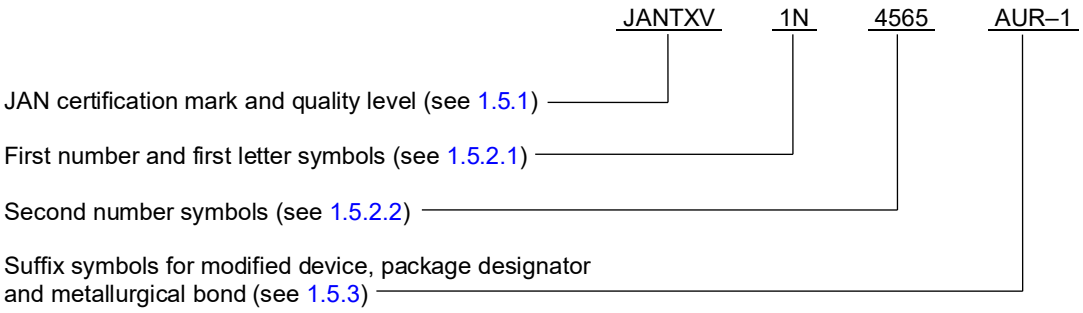
c. Lead finish (see 3.4.1).

d. The complete PIN, see 1.5 and 6.4.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Land and Maritime, VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://qpldocs.dla.mil.

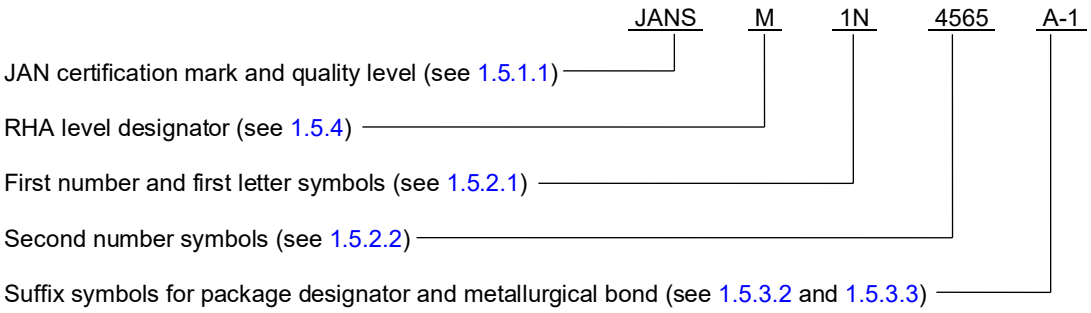
6.4 PIN construction example. The PINs for encapsulated and unencapsulated devices and are constructed using the following forms.

6.4.1 Non-RHA encapsulated devices The PINs for encapsulated devices are constructed using the following form.

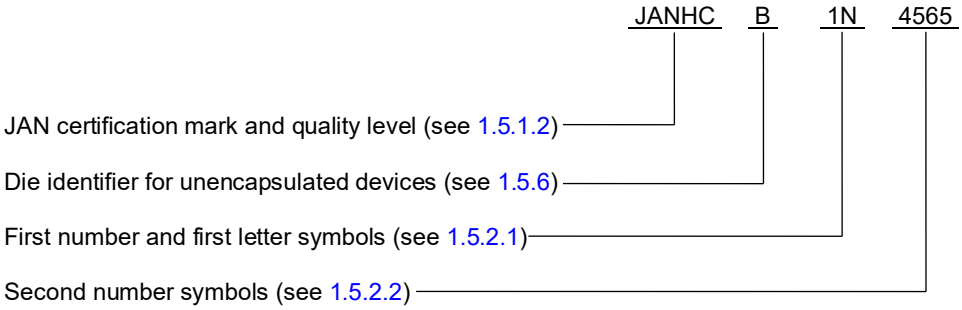


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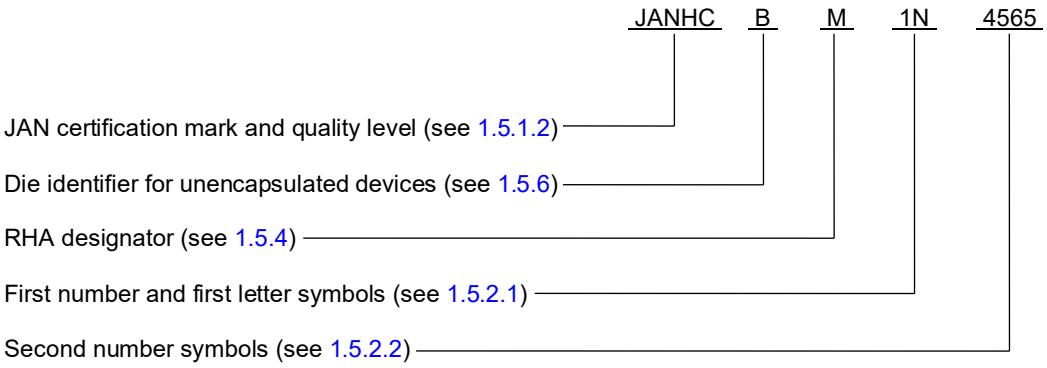
6.4.2 RHA devices. The PINs for RHA devices are constructed using the following form.



6.4.3 Unencapsulated die. The PINs for un-encapsulated devices are constructed using the following form.



6.4.4 RHA unencapsulated die. The PINs for RHA un-encapsulated devices are constructed using the following form.



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6.5 List of PINs.

6.5.1 Encapsulated devices. The following is a list of possible PINs available for encapsulated devices covered by this specification sheet.

PINs for devices in a axial package (1)			
Quality level "JAN"	Quality level "JANTX"	Quality level "JANTXV"	Quality level "JANS"
JAN1N4565A-1	JANTX1N4565A-1	JANTXV1N4565A-1	JANS1N4565A-1
JAN1N4566A-1	JANTX1N4566A-1	JANTXV1N4566A-1	JANS1N4566A-1
JAN1N4567A-1	JANTX1N4567A-1	JANTXV1N4567A-1	JANS1N4567A-1
JAN1N4568A-1	JANTX1N4568A-1	JANTXV1N4568A-1	JANS1N4568A-1
JAN1N4569A-1	JANTX1N4569A-1	JANTXV1N4569A-1	JANS1N4569A-1
JAN1N4570A-1	JANTX1N4570A-1	JANTXV1N4570A-1	JANS1N4570A-1
JAN1N4571A-1	JANTX1N4571A-1	JANTXV1N4571A-1	JANS1N4571A-1
JAN1N4572A-1	JANTX1N4572A-1	JANTXV1N4572A-1	JANS1N4572A-1
JAN1N4573A-1	JANTX1N4573A-1	JANTXV1N4573A-1	JANS1N4573A-1
JAN1N4574A-1	JANTX1N4574A-1	JANTXV1N4574A-1	JANS1N4574A-1
JAN1N4575A-1	JANTX1N4575A-1	JANTXV1N4575A-1	JANS1N4575A-1
JAN1N4576A-1	JANTX1N4576A-1	JANTXV1N4576A-1	JANS1N4576A-1
JAN1N4577A-1	JANTX1N4577A-1	JANTXV1N4577A-1	JANS1N4577A-1
JAN1N4578A-1	JANTX1N4578A-1	JANTXV1N4578A-1	JANS1N4578A-1
JAN1N4579A-1	JANTX1N4579A-1	JANTXV1N4579A-1	JANS1N4579A-1
JAN1N4580A-1	JANTX1N4580A-1	JANTXV1N4580A-1	JANS1N4580A-1
JAN1N4581A-1	JANTX1N4581A-1	JANTXV1N4581A-1	JANS1N4581A-1
JAN1N4582A-1	JANTX1N4582A-1	JANTXV1N4582A-1	JANS1N4582A-1
JAN1N4583A-1	JANTX1N4583A-1	JANTXV1N4583A-1	JANS1N4583A-1
JAN1N4584A-1	JANTX1N4584A-1	JANTXV1N4584A-1	JANS1N4584A-1

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6.5.1 Encapsulated devices. – Continued.

PINs for devices in a UR package (1)			
Quality level "JAN"	Quality level "JANTX"	Quality level "JANTXV"	Quality level "JANS"
JAN1N4565AUR-1	JANTX1N4565AUR-1	JANTXV1N4565AUR-1	JANS1N4565AUR-1
JAN1N4566AUR-1	JANTX1N4566AUR-1	JANTXV1N4566AUR-1	JANS1N4566AUR-1
JAN1N4567AUR-1	JANTX1N4567AUR-1	JANTXV1N4567AUR-1	JANS1N4567AUR-1
JAN1N4568AUR-1	JANTX1N4568AUR-1	JANTXV1N4568AUR-1	JANS1N4568AUR-1
JAN1N4569AUR-1	JANTX1N4569AUR-1	JANTXV1N4569AUR-1	JANS1N4569AUR-1
JAN1N4570AUR-1	JANTX1N4570AUR-1	JANTXV1N4570AUR-1	JANS1N4570AUR-1
JAN1N4571AUR-1	JANTX1N4571AUR-1	JANTXV1N4571AUR-1	JANS1N4571AUR-1
JAN1N4572AUR-1	JANTX1N4572AUR-1	JANTXV1N4572AUR-1	JANS1N4572AUR-1
JAN1N4573AUR-1	JANTX1N4573AUR-1	JANTXV1N4573AUR-1	JANS1N4573AUR-1
JAN1N4574AUR-1	JANTX1N4574AUR-1	JANTXV1N4574AUR-1	JANS1N4574AUR-1
JAN1N4575AUR-1	JANTX1N4575AUR-1	JANTXV1N4575AUR-1	JANS1N4575AUR-1
JAN1N4576AUR-1	JANTX1N4576AUR-1	JANTXV1N4576AUR-1	JANS1N4576AUR-1
JAN1N4577AUR-1	JANTX1N4577AUR-1	JANTXV1N4577AUR-1	JANS1N4577AUR-1
JAN1N4578AUR-1	JANTX1N4578AUR-1	JANTXV1N4578AUR-1	JANS1N4578AUR-1
JAN1N4579AUR-1	JANTX1N4579AUR-1	JANTXV1N4579AUR-1	JANS1N4579AUR-1
JAN1N4580AUR-1	JANTX1N4580AUR-1	JANTXV1N4580AUR-1	JANS1N4580AUR-1
JAN1N4581AUR-1	JANTX1N4581AUR-1	JANTXV1N4581AUR-1	JANS1N4581AUR-1
JAN1N4582AUR-1	JANTX1N4582AUR-1	JANTXV1N4582AUR-1	JANS1N4582AUR-1
JAN1N4583AUR-1	JANTX1N4583AUR-1	JANTXV1N4583AUR-1	JANS1N4583AUR-1
JAN1N4584AUR-1	JANTX1N4584AUR-1	JANTXV1N4584AUR-1	JANS1N4584AUR-1

(1) RHA devices will include the applicable RHA level symbol in the PIN (see 1.5.4 and 6.4.2).

6.5.2 Un-encapsulated devices. The following is a list of possible PINs available for unencapsulated devices covered by this specification sheet.

PINs for quality level "JANHC" (1)			
Die version "A"	Die version "B"	Die version "C"	Die version "D"
JANHCA1N4565A	JANHCB1N4565A	JANHCC1N4565A	JANHCD1N4565A
JANHCA1N4566A	JANHCB1N4566A	JANHCC1N4566A	JANHCD1N4566A
JANHCA1N4567A	JANHCB1N4567A	JANHCC1N4567A	JANHCD1N4567A
JANHCA1N4568A	JANHCB1N4568A	JANHCC1N4568A	JANHCD1N4568A
JANHCA1N4569A	JANHCB1N4569A	JANHCC1N4569A	JANHCD1N4569A
JANHCA1N4570A	JANHCB1N4570A	JANHCC1N4570A	JANHCD1N4570A
JANHCA1N4571A	JANHCB1N4571A	JANHCC1N4571A	JANHCD1N4571A
JANHCA1N4572A	JANHCB1N4572A	JANHCC1N4572A	JANHCD1N4572A

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6.5.2 Un-encapsulated devices. – Continued.

PINs for quality level "JANHC" (1)			
Die version "A"	Die version "B"	Die version "C"	Die version "D"
JANHCA1N4573A	JANHCB1N4573A	JANHCC1N4573A	JANHCD1N4573A
JANHCA1N4574A	JANHCB1N4574A	JANHCC1N4574A	JANHCD1N4574A
JANHCA1N4575A	JANHCB1N4575A	JANHCC1N4575A	JANHCD1N4575A
JANHCA1N4576A	JANHCB1N4576A	JANHCC1N4576A	JANHCD1N4576A
JANHCA1N4577A	JANHCB1N4577A	JANHCC1N4577A	JANHCD1N4577A
JANHCA1N4578A	JANHCB1N4578A	JANHCC1N4578A	JANHCD1N4578A
JANHCA1N4579A	JANHCB1N4579A	JANHCC1N4579A	JANHCD1N4579A
JANHCA1N4580A	JANHCB1N4580A	JANHCC1N4580A	JANHCD1N4580A
JANHCA1N4581A	JANHCB1N4581A	JANHCC1N4581A	JANHCD1N4581A
JANHCA1N4582A	JANHCB1N4582A	JANHCC1N4582A	JANHCD1N4582A
JANHCA1N4583A	JANHCB1N4583A	JANHCC1N4583A	JANHCD1N4583A
JANHCA1N4584A	JANHCB1N4584A	JANHCC1N4584A	JANHCD1N4584A
PINs for quality level "JANKC" (1)			
Die version "A"	Die version "B"	Die version "C"	Die version "D"
JANKCA1N4565A	JANKCB1N4565A	JANKCC1N4565A	JANKCD1N4565A
JANKCA1N4566A	JANKCB1N4566A	JANKCC1N4566A	JANKCD1N4566A
JANKCA1N4567A	JANKCB1N4567A	JANKCC1N4567A	JANKCD1N4567A
JANKCA1N4568A	JANKCB1N4568A	JANKCC1N4568A	JANKCD1N4568A
JANKCA1N4569A	JANKCB1N4569A	JANKCC1N4569A	JANKCD1N4569A
JANKCA1N4570A	JANKCB1N4570A	JANKCC1N4570A	JANKCD1N4570A
JANKCA1N4571A	JANKCB1N4571A	JANKCC1N4571A	JANKCD1N4571A
JANKCA1N4572A	JANKCB1N4572A	JANKCC1N4572A	JANKCD1N4572A
JANKCA1N4573A	JANKCB1N4573A	JANKCC1N4573A	JANKCD1N4573A
JANKCA1N4574A	JANKCB1N4574A	JANKCC1N4574A	JANKCD1N4574A
JANKCA1N4575A	JANKCB1N4575A	JANKCC1N4575A	JANKCD1N4575A
JANKCA1N4576A	JANKCB1N4576A	JANKCC1N4576A	JANKCD1N4576A
JANKCA1N4577A	JANKCB1N4577A	JANKCC1N4577A	JANKCD1N4577A
JANKCA1N4578A	JANKCB1N4578A	JANKCC1N4578A	JANKCD1N4578A
JANKCA1N4579A	JANKCB1N4579A	JANKCC1N4579A	JANKCD1N4579A
JANKCA1N4580A	JANKCB1N4580A	JANKCC1N4580A	JANKCD1N4580A
JANKCA1N4581A	JANKCB1N4581A	JANKCC1N4581A	JANKCD1N4581A
JANKCA1N4582A	JANKCB1N4582A	JANKCC1N4582A	JANKCD1N4582A
JANKCA1N4583A	JANKCB1N4583A	JANKCC1N4583A	JANKCD1N4583A
JANKCA1N4584A	JANKCB1N4584A	JANKCC1N4584A	JANKCD1N4584A

(1) RHA devices will include the applicable RHA level symbol in the PIN (see 1.5.4 and 6.4.4).

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6.6 Substitution information.

6.6.1 Substitution of radiation hardened devices. See MIL-PRF-19500.

6.6.2 Substitutability of dash-one (-1) devices. The non-dash-one devices that were in previous revisions of this document have been deleted from this specification as of MIL PR 19500/452D, dated 1 October 1997. The dash-one devices are a direct substitute for non dash-one devices and are preferred.

6.6.3 Substitution of ΔV_Z devices. The device types within this series with higher second number symbols (lower ΔV_Z voltage-temperature stability) are a direct one-way substitution for lower second number symbols (higher ΔV_Z voltage-temperature stability).

6.6.4 Superseded PINs. The following supersession data applies to PINs associated with this document:

Superseding PIN as specified within MIL-PR-19500/452D, dated 1 October 1997	Superseded PIN as specified within MIL-S-19500/452C, 2 August 1993	Superseding PIN as specified within MIL-PR-19500/452D, dated 1 October 1997	Superseded PIN as specified within MIL-S-19500/452C, 2 August 1993
1N4565A-1	1N4565A	1N4575A-1	1N4575A
1N4566A-1	1N4566A	1N4576A-1	1N4576A
1N4567A-1	1N4567A	1N4577A-1	1N4577A
1N4568A-1	1N4568A	1N4578A-1	1N4578A
1N4569A-1	1N4569A	1N4579A-1	1N4579A
1N4570A-1	1N4570A	1N4580A-1	1N4580A
1N4571A-1	1N4571A	1N4581A-1	1N4581A
1N4572A-1	1N4572A	1N4582A-1	1N4582A
1N4573A-1	1N4573A	1N4583A-1	1N4583A
1N4574A-1	1N4574A	1N4584A-1	1N4584A

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6.7 Suppliers of JANHC and JANKC die. The qualified JANHC and JANKC suppliers with the applicable letter version (example, JANHCAM1N4565A) will be identified on the QML.

JANC ordering information				
Device type	Manufacturer			
	43611	12954	12954	52GC4
1N4565A through 1N4584A 1N4565A through 1N4584A	JANHCA1N4565A through JANHCA1N4584A JANKCA1N4565A through JANKCA1N4584A	JANHCB1N4565A through JANHCB1N4584A JANKCB1N4565A through JANKCB1N4584A	JANHCC1N4565A through JANHCC1N4584A JANKCC1N4565A through JANKCC1N4584A	JANHCD1N4565A through JANHCD1N4584A JANKCD1N4565A through JANKCD1N4584A
	43611 RHA designators M, D, L, R, F, G, H	12954 RHA designators M, D, L, R, F, G, H		52GC4 RHA designators M, D, L, R, F, G, H
1N4565A through 1N4584A 1N4565A through 1N4584A	JANHCA1N4565A through JANHCA1N4584A JANKCA1N4565A through JANKCA1N4584A	JANHCB1N4565A through JANHCB1N4584A JANKCB1N4565A through JANKCB1N4584A		JANHCD1N4565A through JANHCD1N4584A JANKCD1N4565A through JANKCD1N4584A

6.8 Request for new types and configurations. Requests for new device types or configurations for inclusions in this specification sheet should be submitted to: DLA Land and Maritime, ATTN: VAC, Post Office Box 3990, Columbus, OH 43218-3990 or by electronic mail at Semiconductor@dla.mil or by facsimile (614) 693–1642 or DSN 850–6939.

6.9 Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Custodians:
Army - CR
Navy - SH
Air Force - 85
NASA - NA
DLA - CC

Review activities:
Army - AR, MI, SM
Navy - AS, MC
Air Force - 19

Preparing activity:
DLA - CC

(Project 5961-2024-010)

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